[TEGMARK] Charged-lepton module (task #37) — muon as the burden-flip worked example + two blind tier bars

Grace | 2026-07-29 | my 29d lanes: (1) fold the muon into the Forcing+Evidence standard as the worked example of burden-flip tiering; (2) pre-register the two tier rulings I make when the computations land; (3) audit the paper's tiers when Lyra drafts. Ledger: const_010 muon → "DERIVED (e=n without counterexample)" (29d named-ground call, backup .bak.20260729d_muon_burdenflip).

1. Muon as the worked example of BURDEN-FLIP tiering (drop-in for the standard, @Keeper to integrate)

The standard (Part 2, Rule 8 / the K962 ladder) states the burden-flip — "geometric forcing on one route absent a counterexample = DERIVED by default; demotion needs POSITIVE evidence" — but has no worked case for it. The muon is that case, and it is unusually clean because the forced part and the absence-carried part are cleanly separable.

> Worked case (burden-flip, Rule 8 companion). The muon mass ratio $(24/\pi^2)^6$ decomposes into pieces that are *positively derived* and one piece *carried by absence-of-counterexample*, and the honest tag names exactly which is which: - Positively DERIVED: the seat (interior idempotent, ℓ=1, rigorous rank-2 cap); the analytic value 24 = Γ(5) of the domain's Gindikin Γ_Ω (F157=K923, a theorem — Rule 7); the exponent 6 = dim SO(4) forced by π-counting (F111). Four of five criteria are *positive forcings*, not absences. - Absence-carried (the burden-flip step): e = n — that the electroweak SO(4) is the vector-stabilizer of the Jordan identity e. Cal §127 *separated* this from BST's deepest open problem (it is a local Sp(2)-spinor question, not the dim-11→12 gauge embedding); the identification is geometry-FAVORED; the only alternative (n≠e, a conjugate Spin(4) not fixing e) has ZERO positive support. Under the burden-flip this is Derived — but the tag must carry "(e=n without counterexample)," NEVER bare Derived. - Why this is honest, not a wave-through: the qualifier keeps the one absence-carried step *visible*. A bare "Derived" would hide it; "Derived (e=n without counterexample)" shows a hostile reader exactly where the weight rests. Upgrade: Cal's n=e derivation from the substrate potential (SSB-correct — "the unbroken axis n sits at e, breaking transverse," NOT "symmetric wins") removes the qualifier → bare Derived. Demotion: any positive n≠e counterexample → back to Identified. *Lesson: the burden-flip is legitimate only with the absence-carried step named in the tag; an unqualified burden-flip Derived is assertion-drift.*

This is the module's headline discipline: the muon is Derived, honestly, *and* the reader can see the single hinge.

2. Two blind tier bars — pre-registered before the computations land (Cal #27, no wave-through)

Bar A — τ's 71 (Elie forward/blind harness on Lyra's canonical Berezin ν=0 δ_0-norm)

The boundary-mode STRUCTURE is already proven (Γ(0) pole, K982) — that is the derived-final FLOOR and does not move. At stake is only the VALUE (the base 71 / the 49·71 form): - canonical δ_0-norm produces the base FORWARD/BLIND (normalization also canonical, target 3477.48 QUARANTINED) → DERIVED (fixed geometrically, no closed form) — "Derived" = forced by a canonical geometric norm, consistent with having no closed-form algebraic formula (the pole is *why*). - canonical δ_0-norm is a real object and the base is identified with it, but the normalization must be MATCHED (not forced) → IDENTIFIED (real geometric object, matched normalization). - no canonical object / the base does not fall out even against a real δ_0-norm → stays FITTED (derived-final) — this morning's floor holds. - Quarantine: a backward "the δ_0-weight can be chosen to give 71" clears NOTHING — same bar the blind-71 forward already failed once.

Bar B — the count's Value-3 (Lyra + Cal; E7-uniformity is the control)

Ceiling (≤3) stays DERIVED (three routes); interior-seats = 2 stays RIGOROUS. Only Value(=3) is at stake (Selector-1, deliverables A+B): - "each stratum hosts exactly one generation" is FORCED (lower bound: every seat carries a normalizable mode; injectivity: one per seat — target-innocent, electron falling out at the bottom not anchored on m_e) AND the E7-uniformity control gives 4 (the *identical* mechanism yields 3 interior + 1 boundary = 4 on rank-3 E7, i.e. it is a uniform functor not tuned to return 3) → Value-3 DERIVED. - occupancy asserted / matched, OR the mechanism does not uniformly give rank+1 on E7 (returns 3 only by a domain-specific choice) → Value-3 IDENTIFIED. - The E7 control is the anti-tuning guard: a mechanism that gives 3 on D_IV^5 but not 4 on E7 by the same rule is tuned, not forced — that is the whole point of Cal's uniformity check.

3. Paper-tier audit (staged)

When Lyra drafts "three leptons = three strata, one geometry," I audit every tier against this note + the standard before external-ready: muon ALWAYS "(e=n without counterexample)" never bare; electron $6\pi^5$ and muon $(24/\pi^2)^6$ Derived; tau home-Derived (boundary mode) with its value at whatever Bar A returns; count at whatever Bar B returns. Nothing external without Casey's GO.

ADDENDUM — 2026-07-29e: BOTH BARS RULED (K987), banked to the ledgers

Both computations landed on the IDENTIFIED branches of my pre-registered bars. Banked: - Bar A — TAU value → IDENTIFIED (const_039, k987_note; supersedes k983_pending; backup .bak.20260729e_K987). Elie 4910 blind on Lyra F726: canonical vertex weight = Γ_Ω residue Res_0=37.22 ($\sqrt{\pi}$-ful, a=3-odd fingerprint, blind, four digits); base 3479 is O(3479) vs residue O(37) — two orders apart, 71 is a prime with no Γ_Ω origin (closest blind candidate 1824, none within 5%). Home DERIVED (pole), $\sqrt{\pi}$ canonical+blind, base 49·71 MATCHED → value IDENTIFIED. *Honest delta on my own bar:* my Bar-A branch-3 read "base doesn't fall out canonically → FITTED floor"; K987 correctly rules IDENTIFIED (branch 2) because the canonical $\sqrt{\pi}$ + Derived pole home are a real partial geometric object — the base alone is Fitted-flavored, the value as a whole is Identified. I accept the upgrade; it's two-directional (don't under-claim when real canonical structure is present). One last shot = formal-degree scale d(ν); lean Identified-final (prime-71 obstruction stands). - Bar B — count Value-3 → IDENTIFIED (geom_invariants "Number of generations" + "Lepton generations": corrected D→I over-claim AND stale mechanism "N_c=3"→"rank+1=3"; k987_note; backup .bak.20260729e_K987). ★ E7-uniformity PASSED (Cal §132: #strata=r+1 by EJA spectral theorem, D_IV^5→3, E7→4 identical rule) — the anti-tuning conjunct is confirmed, "rigged to 3" is dead. Only one-per-stratum (A+B) remains → Value-3 → DERIVED the moment A+B land, because the E7-control already licenses Derived-over-coincidence. - @Keeper: the standard's Part 4 Selector-1 line reads "Value(=3) REDUCED to A+B" — update to "IDENTIFIED; E7-uniformity PASSED (Cal §132, banked); one-per-stratum A+B the only open piece." And the DERIVED-list can add the muon (K986, "Derived (e=n without counterexample)"). - Quark handle (my radar for when they open): do NOT transplant the lepton boundary-strata method — K768 (rank-1) points to quarks as bulk modes (hierarchy = off-rank-1 spectrum of one condensate), not three idempotent seats. Corpus-reconnect first ("how do quarks actually sit?") before any method transplant.

— Grace, 2026-07-29 [TEGMARK]. Muon = the burden-flip worked example (forced pieces + one absence-carried e=n step, tag mandatory-qualified); ledger const_010 → DERIVED (e=n without counterexample). Two blind bars: τ's 71 (canonical δ_0-norm forward/blind → Derived-geometric; matched-normalization → Identified; no object → Fitted-floor) and count Value-3 (forced occupancy + E7-uniformity-gives-4 → Derived; asserted/non-uniform → Identified). @Keeper integrate the worked case into the standard's Rule-8/burden-flip; I rule both tiers as they land and audit the paper.